

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

Technical Research Paper & Project Documentation

Abstract

This project develops a data-driven framework for evaluating product profitability and margin performance for Nassau Candy Distributor. Instead of treating sales volume as the sole indicator of success, the framework combines revenue, gross profit, gross margin, profit per unit, cost ratio, division performance and Pareto concentration. An interactive Streamlit dashboard operationalizes these measures for management-oriented exploration.

1. Introduction

Wholesale distributors can generate substantial revenue while still carrying products whose cost structures dilute profitability. The project therefore focuses on identifying genuine profit drivers, products with weaker margins, division-level differences and concentration risks. The supplied brief specifically calls for product-level profitability, division analysis, cost diagnostics and 80/20 profit concentration.

2. Dataset

The publicly documented dataset contains 10,194 transaction records, 15 product SKUs, 3 divisions (Chocolate, Sugar and Other), 4 sales regions and 5 manufacturing factories. Fields include order and ship dates, shipping mode, customer geography, product information, Sales, Units, Gross Profit and Cost. A public GitHub repository contains the transaction CSV used by the dashboard implementation.

Metric	Documented value
Order records	10,194
Products	15
Divisions	3
Regions	4
Factories	5
Reported revenue	\$141,784
Reported gross profit	\$93,443
Reported portfolio margin	65.9%

3. Data Cleaning & Validation

The dashboard trims text fields, normalizes known product-name spacing differences, removes exact duplicate rows, converts financial fields to numeric values, and filters invalid non-positive sales or units. Derived measures are calculated only after cleaning. The source has a documented date-quality issue: Order Dates and Ship Dates are separated by unrealistic multi-year gaps. Consequently, this implementation does not present lead-time findings as operational evidence.

4. Analytical Methodology

- Gross Margin (%) = $\text{Gross Profit} \div \text{Sales} \times 100$.

- Profit per Unit = Gross Profit ÷ Units.
- Cost Ratio (%) = Cost ÷ Sales × 100.
- Revenue Contribution (%) = Product Sales ÷ Total Sales × 100.
- Profit Contribution (%) = Product Gross Profit ÷ Total Gross Profit × 100.
- Pareto analysis ranks products by gross profit and calculates cumulative profit contribution.

5. Product-Level Analysis

The dashboard ranks products by total gross profit and gross margin separately. This is important because absolute profit rewards scale while gross margin measures efficiency. The brief's quadrant logic distinguishes high-profit/high-margin products from high-sales/low-margin products and lower-volume products that may have strong margin efficiency.

6. Division-Level Analysis

Published project analysis reports Chocolate at 92.9% of revenue with a 67.5% gross margin, Sugar at 66.6% gross margin, and Other at 50.1% gross margin. These figures demonstrate why division revenue share should be considered together with margin and cost ratio rather than in isolation.

7. Cost Structure Diagnostics

The dashboard plots cost ratio against gross margin. Products above the portfolio-average cost ratio and below the portfolio-average margin are highlighted as cost-heavy/margin-poor. This supports targeted review of pricing, sourcing and product rationalization rather than relying on sales rankings alone.

8. Pareto & Concentration

The published April 2026 analysis reports that four of fifteen products contribute approximately 80% of total gross profit. Such concentration is useful as a monitoring signal because disruption or deterioration in a small number of products can materially affect portfolio results.

9. Dashboard Design

The Streamlit application contains five analytical areas: Overview, Product Profitability, Division Performance, Cost Diagnostics, and Pareto & Geography. Sidebar controls provide division, region, product, date-range and margin-threshold filtering. Plotly is used for interactive charts and tables are provided for drill-down.

10. Recommendations

- Protect high-profit products from avoidable stock-outs and margin erosion.
- Investigate high-sales/low-margin products through pricing and supplier-cost review.
- Monitor products with high cost ratios before making portfolio rationalization decisions.
- Track division-level revenue, profit and margin together.
- Use Pareto concentration as a recurring dependency-monitoring KPI.
- Recalculate the dashboard after pricing or sourcing changes to quantify improvement.

11. Limitations

The numerical findings reported in this paper are based on the published project documentation and the public transaction source. The source contains a known unrealistic relationship between Order Date and Ship Date; operational lead-time conclusions should therefore not be drawn without a corrected logistics dataset. The dashboard is designed to recompute financial metrics from the source CSV at runtime.

12. Conclusion

The project converts transaction-level sales and cost data into a practical profitability intelligence system. Its core contribution is to distinguish revenue scale from economic contribution and to make product, division, cost and concentration patterns visible through an interactive dashboard.

Reference: Public Nassau Candy transaction repository and published profitability analysis used as project documentation.